# AI announcement

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## AI announcement

In my assignment1

I used ChatGPT as my support tool to help me understand some functions and grammar that I didn't know. The AI provided some code examples to help me understand the functions that I hadn't learned. I used AI to: Show me how to use the functions that I want to code with my assignment. provide some example to read all csv data in one folder. explain some functions grammar and check the correctness of my code. The completion of this assignment and the code writing were all done by myself. I only used the AI to tell me how to understand the function packages and the grammar logic of the functions, and pointed out the problems in the code that I wrote.

## Screenshots

ChatGPT

how can I use lubridate package to convert the date values from character to date format? such as convert 22/09/2024 and Mar 20, 2025 to date format? in short answer

Use the lubridate functions based on the format:

```r
library(lubridate)

dmy("22/09/2024")     # "2024-09-22"
mdy("Mar 20, 2025")  # "2025-03-20"
```

For mixed formats:

```r
parse_date_time(c("22/09/2024", "Mar 20, 2025"),
                orders = c("dmy", "mdy"))
```

ChatGPT

how to read the boxplot in R with different vector ?

To read a **boxplot in R with different vectors**, you mainly compare their distributions.

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### ✅ Example (multiple vectors)

```r
a <- c(1, 2, 3, 4, 5)
b <- c(2, 3, 4, 5, 6)

boxplot(a, b, names = c("A", "B"))
```

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### 📊 How to interpret the boxplot

For each box (vector):

- **Middle line** → median (center value)
- **Box (IQR)** → middle 50% of data (Q1 to Q3)
- **Whiskers** → range (excluding outliers)
- **Points outside** → outliers

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### 🔍 Comparing different vectors

- **Higher box** → larger values overall
- **Taller box** → more variability
- **Median position** → skewness
- **Outliers** → unusual values

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👉 In short:
A boxplot lets you quickly compare **center (median), spread (IQR), and outliers** across multiple vectors.

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ChatGPT

if I want to add some details to improve my ggplot such as colors and width and height, how can I write the code? in short answer

Use `ggplot2` layers and saving options:

```r
library(ggplot2)

ggplot(df, aes(x, y, fill = group)) +
  geom_boxplot(width = 0.5) +        # control box width
  scale_fill_manual(values = c("red", "blue")) +  # colors
  theme_minimal()
```

Set plot size when saving:

```r
ggsave("plot.png", width = 6, height = 4)
```

👉 `fill` = colors, `width` = box width, `ggsave()` = figure size

ChatGPT

how to inline the coefficient after i summary the linear regression? such as the r squared and p-value?

In **R Markdown**, you can inline results from a linear model using `` `r ...` ``.

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### ✅ Example

```r
model <- lm(y ~ x, data = df)
```

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### 📌 Inline $R^2$ and p-value

```markdown
R-squared is `r summary(model)$r.squared`
```

```markdown
p-value is `r summary(model)$coefficients[2, 4]`
```

---

### ✨ Optional (format nicely)

```markdown
R-squared is `r round(summary(model)$r.squared, 3)`
```

```markdown
p-value is `r signif(summary(model)$coefficients[2, 4], 3)`
```

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👉 In short:

- Use `` `r ...` `` for inline code
- `summary(model)$r.squared` → $R^2$
- `summary(model)$coefficients[row, col]` → p-value

ChatGPT

Shar

if i want to put the screenshot in the pmd, which code can I use and I also want to add some commit and words.

In a `.Rmd` **(R Markdown)** file, you can insert screenshots and add comments/text like this:

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### ✅ 1. Insert an image (screenshot)

Markdown

```
![My screenshot](images/screenshot.png)
```

👉 Make sure the image is in your project folder (e.g., `images/`)

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### ✅ 2. Add comments / explanation text

Just write normally in Markdown:

Markdown

```
This plot shows the distribution of values across groups. The median is higher in group A
```

Ask anything

ChatGPT can make mistakes. Check important info.

ChatGPT

Sha

### ✅ 3. Add code + explanation together

R

```r
# This code creates a boxplot
ggplot(df, aes(x, y)) +
  geom_boxplot()
```

Then below:

Markdown

```
The boxplot indicates variability and outliers in the dataset.
```

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### ✅ 4. Optional: resize image

Markdown

```
<img src="images/screenshot.png" width="400">
```

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👉 In short:

- Use `![ ]( )` → insert image
- Write text directly → comments/explanation

Ask anything